7. A technician sets up the vending machine for use

The technician initializes the controller and checks that the machine is ready for customers.

Objective	• Initialize the controller, load the starting prices, and prepare the machine for normal operation.	
Results	• All 16 locations have their assigned default prices, and the controller begins with no active purchase or deposited balance.	
Actors	• Technician, who understands board programming and the reset procedure; vending machine controller.	
Related use cases	• Use Case 1: Buying an item. Use Case 8: Changing an item's price.	
Trigger	• The technician prepares the controller for use and presses KEY0 to simulate a power-up reset.	
Stakeholders	• The owner and customers, who depend on the controller starting correctly.	
Resources	• Programmed SoC Series V board, and Quartus.	
Assumptions	• A working setup with the seven-segment display, working money switches, working RAM set up for Items and the working keypad as well.	
Post-condition	• The controller is ready for a new customer, with default prices loaded and the payment balance at $0.00.	
Frequency	• During initial setup and whenever a power-up reset is performed.	
Mission Priority	• High. Normal purchases depend on correct initialization.	
Safety Priority	• Should hold no safety concern	
Flow of events	**Step #**	**Action**
	1.	The technician powers and programs the board as needed, then places SW0 through SW3 in the off position.
	2.	The technician presses KEY0.
	3.	The controller loads the default prices for all 16 valid locations and clears the current payment, selection, and previous change indication.

	4.	The technician releases KEY0, and the controller enters its normal ready state.
	5.	The technician selects each valid location and checks that its displayed price matches the default price table.
	6.	After checking the prices, the technician presses and releases KEY0 again to leave the controller ready with no active selection or deposited money.
Alternate Events	**Step #**	**Action**
	1A	If the board has not been programmed, the technician loads the controller design before continuing.
	2A	If prices were previously changed in maintenance mode, the reset reloads the default prices.
	5A	If a location displays an incorrect price, the technician checks the initialization data, corrects the problem, and repeats the setup checks.
Hazard/ Result /Mitigation	• If case is used incorrectly, the entire system is likely incorrect or not working to specifications.	
Requirements from Use Case	• Use KEY0 to perform the simulated power-up reset. • Load default prices for locations 11, 12, 13, 14, 21, 22, 23, 24, 31, 32, 33, 34, 41, 42, 43, and 44. • Keep every default price within $0.05 through $9.99. • Initialize the customer payment balance to zero and clear and clear any active transaction. • Enter normal operation after initialization and display the correct price for each valid selection.	